

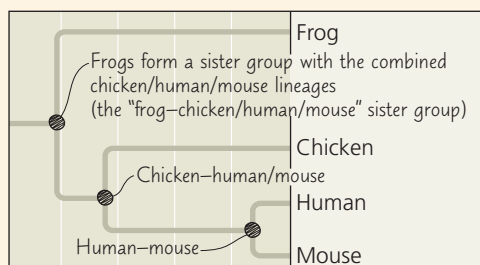
26 Phylogeny and the Tree of Life

KEY CONCEPTS

- 26.1** Phylogenies show evolutionary relationships p. 554
- 26.2** Phylogenies are inferred from morphological and molecular data p. 558
- 26.3** Shared characters are used to construct phylogenetic trees p. 559
- 26.4** An organism's evolutionary history is documented in its genome p. 565
- 26.5** Molecular clocks help track evolutionary time p. 566
- 26.6** Our understanding of the tree of life continues to change based on new data p. 568

Study Tip

Label a drawing: The members of a *sister group* are each other's closest relatives, making sister groups a useful way to describe evolutionary relationships. As in this example, identify the sister groups associated with each branch point in Figures 26.11, 26.12, 26.14, and 26.16.



➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 26
- Figure 26.12 Walkthrough: Using Derived Characters to Infer Phylogeny

For Instructors to Assign (in Item Library)

- Misconception Questions 126, 128, 129
- Tutorial: Visualizing Phylogenetic Relationships

Ready-to-Go Teaching Module

(in Instructor Resources)

- Phylogenetic Trees (Concept 26.3)



Figure 26.1 Although this animal resembles a snake, it is actually a legless lizard: the glass lizard *Ophisaurus apodus*. Why isn't the glass lizard considered a snake? Biologists based this decision in part on a comparison of its traits with those of snakes—and the glass lizard does not have a highly mobile jaw or many vertebrae, two traits shared by all snakes.

How do biologists distinguish and categorize the millions of species on Earth?

Traits shared due to **common ancestry** are used to classify organisms into groups that reflect their **evolutionary history**:

This branch point represents the **common ancestor** of all the groups shown here.

